



CENG 407

Innovative System Design and Development I
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**MULTI-LABEL CHEST X-RAY DISEASE
CLASSIFICATION AND EXPLAINABILITY WITH
DEEP LEARNING**

LITERATURE REVIEW

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1. Model Structure Overview

All models in this project share the same backbone and architecture, ensuring a consistent baseline for evaluation.

- **Backbone Architecture:** DenseNet-121
- **Attention:** Convolutional Block Attention Module (CBAM)
- **Input Resolution:** 512×512 pixels downscaled from 1024x1024
- **Output Layer:** 14 class multi label classification

1.1. DenseNet-121 Backbone

DenseNet-121 was chosen for its proven success in medical image classification. Its dense connectivity passes outputs from each layer to all subsequent layers, reducing vanishing gradient issues. Pretrained ImageNet weights also allow faster and more stable training and the modular design of DenseNet makes it easy to add attention modules like CBAM.

1.2. CBAM (Convolutional Block Attention Module)

CBAM is an attention mechanism that refines feature maps along both channel and spatial dimensions.

- **Channel Attention:** Determines which features are important by applying average and max pooling across the spatial dimensions, passing the results through a small MLP, and multiplying the resulting attention map with the original features.
- **Spatial Attention:** Identifies important regions within the feature map by applying pooling across channels, concatenating the results, passing them through a convolution, and multiplying the resulting map with the original features.

By adding CBAM after each dense block, the models focus on important regions of the chest X-ray. This helps the model detect rare thoracic illnesses that might be missed by standard convolutional layers.

2. Model 1: Baseline BCEWithLogitsLoss:

At the beginning of the project, a baseline model was created utilizing the BCEWithLogitsLoss function. It is considered the industry standard for multi-label medical image classification. The main objective of this model was to understand the distribution of the dataset and set a benchmark for future improvements.

The baseline model was trained using the following core hyperparameters and configurations:

- **Backbone Architecture:** DenseNet-121 integrated with CBAM (Convolutional Block Attention Module)
- **Input Resolution:** 512x512 (Standardized and Raw Images)
- **Optimization:** Adam Optimizer, Learning Rate (LR) = $1e-4$, Batch Size = 8
- **Decision Mechanism:** Sigmoid activation followed by a fixed threshold of 0.5 for all classes.

The dataset was splitted into 80% Training, 10% Validation, and 10% Testing. The training pipeline was structured so that the validation set was evaluated at the end of every training epoch. This allowed for continious monitoring of the model's learning curve. Also it gave an opportunity for early detection of overfitting.

2.1. Training and Validation Phase:

During training, the model learned very fast. It focused on the "healthy" pixels and the most common disease labels. According to the training logs, the results for Training and Validation finally became stable at these levels:

Metric	Value
Macro F1	0.2612

Note: Throughout the training process, it became clear that the BCEWithLogitsLoss unintentionally encouraged the model to predict "Negative" (the absence of disease). This was used as the primary mathematical strategy to minimize the overall loss

2.2. Testing Phase and Performance Evaluation:

The trained model was then evaluated on the unseen 10% Test set to assess its generalization capabilities. The resulting metrics clearly showed the misleading part of the accuracy concept in imbalanced medical datasets.

Disease	AUC Score	F1 Score	Accuracy
Cardiomegaly	0.8259	0.3970	0.9039
Consolidation	0.8955	0.3441	0.9637
Edema	0.8052	0.0930	0.9768
Effusion	0.8153	0.1053	0.9949
Emphysema	0.8811	0.5050	0.9051
Fibrosis	0.8732	0.3838	0.9752
Hernia	0.7650	0.1443	0.9753
Infiltration	0.8997	0.4898	0.9963
Mass	0.6591	0.1556	0.8565
Nodule	0.8570	0.4152	0.9427
Pleural_Thickening	0.7935	0.3360	0.9378
Pneumonia	0.7819	0.1567	0.9600
Pneumothorax	0.6925	0.0312	0.9908
AVERAGE (MACRO)	0.8111	0.2736	0.9522

2.3. In-Depth Analysis and Findings:

A deeper analysis of the results reveals three basic limitations of the baseline approach:

1. **The Accuracy Paradox:** The model has a very high accuracy (95.22%), but the Macro F1 score is very low (0.2736). This happens because of "class imbalance." The model stays accurate just by saying "Healthy" for almost every case. For example, in Pneumothorax, the accuracy is 99.08%, but the model's real ability to find the disease (F1 Score) is almost zero (0.0312).
2. **The Fixed Threshold Problem:** The model uses a fixed 0.5 limit to decide if a disease exists. However, for rare diseases, the model only gives a probability between 0.10 and 0.30. Because 0.5 is too high, the model ignores these signals and misses many sick patients.
3. **Loss Function (BCE) Problem:** The standard BCE loss treats healthy samples and rare disease samples the same way. Since there are millions of "healthy" pixels, the model focuses on them and ignores the small, important details of rare diseases.

2.4. Conclusion for Model 1:

The results prove that a fixed threshold and classical BCE loss are not suitable for medical diagnosis on imbalanced data. To overcome this bottleneck and improve diagnostic reliability, a strategic shift toward an Asymmetric Loss (ASL) architecture is required to force the network to focus on hard-to-detect, rare pathologies.

3. Model 2: Asymmetric Loss (ASL) Implementation:

Following the evidence that the baseline BCEWithLogitsLoss was limited by the "Accuracy Paradox" and class imbalance, the architecture was upgraded to use Asymmetric Loss (ASL). The main goal of this model was to stop the large number of easy-negative (healthy) samples from dominating the training. This forces the network to focus on rare and hard-to-detect diseases instead.

The model retained the core DenseNet-121 + CBAM backbone and 512x512 input resolution, but the loss function was replaced with a custom ASL module configured with the following configurations:

- **Focusing Parameters:** $\gamma_{\text{neg}} = 4$ and $\gamma_{\text{pos}} = 0$. This setup heavily penalizes the network for missing real diseases while reducing the influence of easy negative samples.
- **Asymmetric Clipping:** $\text{clip} = 0.05$. This feature removes the "background noise" from obvious healthy tissues. If the model is already very confident that a sample is healthy ($p < 0.05$), the negative loss is automatically set to zero.
- **Decision Mechanism:** The threshold remained at 0.5 to compare with Model 1.

Just like the baseline model, this version was trained with continuous monitoring of Training/Validation metrics to find the best possible checkpoint.

3.1. Training and Validation Phase:

The training logs showed a much healthier learning curve compared to the baseline model. Instead of reaching a limit by simply predicting zeros(negative), the model steadily improved its ability to extract features. By Epoch 15, the model reached its highest validation score. After this point, the validation loss began to stabilize. It shows that the model had reached its maximum learning potential for this configuration.

Metric	Value
Macro F1	0.3321

3.2. Testing Phase and Performance Evaluation:

The optimal weights were applied to the unseen 10% Test set. The resulting metrics showed a dramatic shift in how the model makes predictions. This change effectively destroyed the accuracy illusion that was seen in Model 1.

Disease	AUC Score	F1 Score	Accuracy
Atelectasis	0.8360	0.4223	0.8598
Cardiomegaly	0.9108	0.3646	0.9648
Consolidation	0.7469	0.1973	0.9649
Edema	0.7446	0.1154	0.9932
Effusion	0.9009	0.5448	0.8768
Emphysema	0.9051	0.5272	0.9755
Fibrosis	0.7782	0.2159	0.9633
Hernia	0.8492	0.5128	0.9972
Infiltration	0.6975	0.3283	0.5690
Mass	0.8797	0.4374	0.9266
Nodule	0.7807	0.3790	0.9070
Pleural_Thickening	0.8013	0.2649	0.9340
Pneumonia	0.7176	0.0444	0.9872
Pneumothorax	0.9043	0.4875	0.9269
AVERAGE (MACRO)	0.8181	0.3458	0.9176

3.3. In-Depth Analysis and Comparative Findings:

Comparing Model 2 against the BCE Baseline shows that changing the architecture is a crucial step for the model to evaluate the X-Ray images.

1. **Breaking the Accuracy Paradox:** The most important change is the decrease in Average Accuracy from 95.22% (Model 1) to 91.76% (Model 2). In this project, this drop is actually a major success. It proves that the ASL implementation stopped the model from simply predicting negative to fake a high accuracy. The model is now making real disease predictions, which is shown by the significant jump in the Macro F1 score from 0.2736 to 0.3458.

2. **Massive Gains in Rare Pathologies:** The power of ASL's hard-negative mining is most visible in the rarest and most difficult diseases:

- **Hernia:** F1 score rose from a very low 0.1443 (Baseline) to an impressive 0.5128.
- **Pneumothorax:** Experienced a massive leap from 0.0312 to 0.4875.
- **Effusion:** Improved drastically from 0.1053 to 0.5448. These leaps confirm that ignoring the gradients of easy negatives successfully allowed the network's attention layers (CBAM) to extract the faint visual signatures of severe, rare diseases.

3. **The Remaining Bottleneck:** Despite the progress, some classes like Pneumonia still have low scores (F1: 0.0444). Although ASL increased the probability scores for sick patients, the code still uses a rigid 0.5 boundary to make decisions:

```
preds = (torch.sigmoid(outputs) > 0.5).float()
```

Because diseases like Pneumonia are very hard to see on a 2D X-ray, their probability scores often stay below 0.5. As a result, the fixed threshold labels them as negative and that causes a wrong evaluation.

3.4. Conclusion for Model 2 (B2 Level)

The switch to Asymmetric Loss (ASL) successfully solved the gradient imbalance problem. This improved the Macro F1 score to 0.3458 and proved to be effective for rare diseases. However, the analysis shows that using a fixed 0.5 threshold prevents the model from reaching its full diagnostic potential. This result clearly points to the next step in the model's development: implementing an adaptive, class-specific threshold strategy to optimize the results for each disease.

4. Model 3: ASL with Image Preprocessing (CLAHE + Gamma Correction):

In an effort to better detect subtle pathological features—such as faint lung patterns or small nodules—an image preprocessing pipeline was introduced for Model 3. The hypothesis was that by enhancing the contrast and standardizing the lighting of the X-rays, the hard-to-detect cases would become easier to identify.

This model utilized the same model (DenseNet-121 + CBAM) and optimization strategy (Asymmetric Loss with $\gamma_{\text{neg}} = 4$, $\text{clip} = 0.05$) as Model 2. However, the input dataset was completely preprocessed using Scenario 1: CLAHE (Contrast Limited Adaptive Histogram Equalization) combined with Gamma Correction.

Just like the previous models, we checked the results at the end of each training step (epoch) to see how well the model was learning.

4.1. Training and Validation Phase:

During training, the validation loss curve for this model was not as good as the first model (the one with raw images). It was a bit unstable. The model reached its best performance at Epoch 15. After this point, the model started to overfit, which means it focused too much on the special details of the cleaned images and stopped learning useful information for new cases.

Metric	Value
Macro F1	0.3208

4.2. Testing Phase and Performance Evaluation

The optimal weights from Epoch 15 were tested on the preprocessed 10% Test set. The resulting metrics provided crucial evidence regarding how deep learning attention mechanisms interact with artificially altered radiographic images.

Disease	AUC Score	F1 Score	Accuracy
Atelectasis	0.8308	0.4167	0.8465
Cardiomegaly	0.8810	0.3415	0.9639
Consolidation	0.7647	0.1550	0.9482
Edema	0.7815	0.0889	0.9939
Effusion	0.8984	0.5027	0.8504

Disease	AUC Score	F1 Score	Accuracy
Emphysema	0.9347	0.5323	0.9731
Fibrosis	0.7810	0.1935	0.9703
Hernia	0.8551	0.5000	0.9973
Infiltration	0.6885	0.3262	0.5772
Mass	0.8835	0.4502	0.9155
Nodule	0.7864	0.3556	0.8982
Pleural_Thickening	0.7805	0.2248	0.9293
Pneumonia	0.6584	0.0571	0.9902
Pneumothorax	0.9062	0.4856	0.9415
AVERAGE (MACRO)	0.8165	0.3307	0.9140

4.3. In-Depth Analysis and Comparative Findings

Comparing the results of the preprocessed model (Model 3) against the raw image model (Model 2) leads to a very important conclusion regarding medical image processing:

1. **Performance Degradation:** The initial hypothesis was proven incorrect. Applying CLAHE and Gamma Correction actually reduced the overall Macro F1 score from 0.3458 (Model 2) to 0.3307 (Model 3). While this is still much better than the first BCE model, it represents a step backward from the best strategy.
2. **Sensitivity of the CBAM Architecture:** The performance drop is primarily attributed to how the Convolutional Block Attention Module (CBAM) processes images. CLAHE fundamentally alters pixel intensity distributions, often introducing artificial noise and harsh boundaries, known as halo artifacts, in areas of soft tissue. The spatial attention mechanisms of the CBAM layers likely locked onto these artificial artifacts rather than the genuine underlying pathological structures. This led to a loss of diagnostic precision in specific classes. For example, the F1 score for Consolidation dropped from 0.1973 to 0.1550, and Effusion dropped from 0.5448 to 0.5027.

3. **Clinical Alignment:** This mathematical finding matches the feedback from real doctors (radiologists). Doctors rely on natural, subtle changes in X-rays to tell the difference between overlapping tissues. Strong image enhancements can destroy these details. By proving that the model performs worse on altered images, we confirm that keeping the original (raw) image quality is better.

4.4. Conclusion for Model 3:

The results show that aggressive image preprocessing (CLAHE + Gamma) is not suitable for this specific DenseNet-CBAM architecture. The model extracts more reliable features from raw images. Consequently, the final version of the model will stop using artificial enhancements and return to Raw 512x512 images. We will now focus entirely on the last problem found in Model 2: the static decision threshold.

5. Model 4: Asymmetric Loss with Dynamic Class-Specific Thresholds (The Final Model)

Based on evidence from previous phases, the final model architecture was solidified. We reverted to Raw Standardized 512x512 Images because Model 3 showed that artificial enhancements degrade the CBAM mechanism. While Model 2 proved that Asymmetric Loss (ASL) solves gradient imbalance, the rigid 0.5 decision boundary remained a bottleneck, suppressing the diagnosis of ambiguous conditions.

To maximize diagnostic potential, we introduced Dynamic Class-Specific Threshold Optimization. This iteration used a specialized learning rate of $5e-5$ and an Adam optimizer. Although 30 epochs were planned, training was stopped at Epoch 15 due to instability in the loss curves and clear signals of overfitting. To ensure the best generalization, the model weights and optimal thresholds from this peak point were saved together in the final checkpoint.

How the Dynamic Thresholds are Calculated: Instead of using a fixed 0.5 cutoff, the system learns unique thresholds for each pathology. After validation, the algorithm extracts raw probability scores and systematically tests values from 0.10 to 0.85 with steps of 0.05. It then selects the exact threshold that maximizes the F1 Score for each specific disease. Consequently, easily confused diseases receive stricter thresholds, while faint, hard-to-detect conditions receive more sensitive ones. This automated process ensures reliable clinical support across all 14 pathologies

5.1. Training and Validation Phase

The model was trained for 30 epochs using the ASL framework, and the training logs show a perfect example of deep learning optimization. The model reached its absolute peak performance at Epoch 10, achieving the highest validation score recorded in the entire project. After this point, the validation loss began to rise while the F1 score stopped improving and eventually declined, dropping to 0.3372 by Epoch 15. This indicated the start of overfitting, where the model begins to memorize the data rather than learning general patterns. Consequently, the weights from Epoch 10 were permanently saved as the best version of the model.

Metric	Value
Val Macro F1	0.3504

5.2. Testing Phase and Performance Evaluation

The weights from Epoch 10 (best model), along with the dynamic thresholds, were applied to the unseen Test set. The resulting table below showcases the optimized performance of the system.

Disease	Threshold	AUC Score	F1 Score	Accuracy
Atelectasis	0.5000	0.8356	0.4204	0.8669
Cardiomegaly	0.4500	0.9137	0.3627	0.9645
Consolidation	0.5000	0.7703	0.1739	0.9577
Edema	0.5500	0.7787	0.1818	0.9947
Effusion	0.5500	0.8980	0.5769	0.9024
Emphysema	0.6500	0.9391	0.5404	0.9780
Fibrosis	0.5000	0.8027	0.2115	0.9490
Hernia	0.7500	0.8653	0.5789	0.9976
Infiltration	0.5000	0.6870	0.3215	0.5781
Mass	0.6500	0.8855	0.4551	0.9306
Nodule	0.5500	0.8038	0.3739	0.9045

Disease	Threshold	AUC Score	F1 Score	Accuracy
Pleural_Thickening	0.5500	0.8015	0.2517	0.9364
Pneumonia	0.3500	0.7196	0.0653	0.9320
Pneumothorax	0.5500	0.9158	0.5151	0.9452
AVERAGE (MACRO)	-	0.8298	0.3592	0.9170

5.3. In-Depth Analysis and Final Comparative Findings

The transition to Model 4 represents a definitive engineering success. By breaking down the metrics and examining the optimized thresholds, several profound conclusions can be drawn about the model's behavior:

1. **The Macro F1 Breakthrough:** The final test Macro F1 score reached 0.3592, with a Macro AUC of 0.8298. This represents a crucial, targeted improvement over the static-threshold ASL approach, which had an F1 of 0.3458 in Model 2. It mathematically proves that while Asymmetric Loss effectively addresses class imbalance, replacing the rigid 0.5 decision boundary with Dynamic Thresholds is the ultimate factor for unlocking the model's full diagnostic potential in medical datasets.
2. **Anatomy of the Thresholds:** Unlike Model 2, which forced a uniform 0.5 decision boundary across all diseases, the calculated dynamic thresholds reveal how the model adapted to the visual characteristics of each pathology:
 - **The Sensitive Approach (Pneumonia - 0.35):** Pneumonia is difficult to isolate on 2D X-rays due to overlapping lung textures. While Model 2's static 0.5 threshold suppressed these faint signals, the algorithm dynamically lowered the cutoff to 0.35. This means the model acts more sensitively; even at a 35% probability, it flags the disease to ensure no potential cases are missed.
 - **The Conservative Approach (Hernia - 0.75):** Conversely, instead of risking false positives at the standard 0.5 mark, the threshold for Hernia was raised to 0.75. Because Hernia features can easily be confused with other anomalies, the model learned to be extremely strict. It only makes a positive prediction when it is absolutely confident, resulting in an exceptional F1 score of 0.5789 for one of the rarest diseases in the dataset.

3. **Final Resolution of the Diagnostic Trade-off:** The final Average Accuracy rests at 91.70%. While statistically identical to the static-threshold ASL model (91.76%), this metric now represents a sharper diagnostic capability. Because of the thresholds are being dynamically optimized, the model successfully extracts more true positive cases without blindly increasing the false positive rate. That ensures model actively and accurately identifies multi-label pathologies.

5.4. Final Conclusion:

The systematic evolution from a standard BCE architecture to a highly tailored Raw Image, DenseNet121-CBAM, ASL, and Dynamic Threshold pipeline has yielded a robust clinical decision-support model. Achieving a Macro F1 score of nearly 0.36 on the NIH ChestX-ray14 dataset places this framework in a highly competitive standing. It closely approaches the established statistical baseline of human inter-radiologist agreement, which traditionally hovers around an F1 of 0.38 for this specific dataset.

6. Comparison of Models:

To provide a better overview of the project's evolution in architecture and the resulting performance gains, the following table summarizes the test metrics across all four model iterations. It shows the changes in AUC score, Macro F1 and Avg. Accuracy.

Model Iteration	Loss Function	Preprocessing	Decision Threshold	Macro AUC	Macro F1	Avg. Accuracy
Model 1 (Baseline)	BCEWithLogits	Raw Images	Static (0.50)	0.8111	0.2736	0.9522
Model 2 (ASL)	Asymmetric Loss	Raw Images	Static (0.50)	0.8181	0.3458	0.9176
Model 3 (Processed)	Asymmetric Loss	CLAHE Gamma +	Static (0.50)	0.8165	0.3307	0.9140
Model 4 (Final Model)	Asymmetric Loss	Raw Images	Dynamic (Optimized)	0.8298	0.3592	0.9170

7. EXPERIMENTS:

The primary objective of these experiments is to evaluate the diagnostic evolution of the developed models under real-world scenarios. By testing all four model iterations on the same set of unseen radiographs from the NIH ChestX-ray14 test split, we aim to:

- **Validate Progression:** Mathematically and visually demonstrate how shifting from BCE to Asymmetric Loss (ASL) improves the detection of rare and overlapping pathologies
- **Assess Preprocessing Impact:** Determine whether raw image data or enhanced imagery, specifically using CLAHE and Gamma Correction, yields higher diagnostic reliability.
- **Verify Threshold Optimization:** Prove that Dynamic Class-Specific Thresholds in Model 4 successfully convert probability scores into accurate True Positive detections that a static 0.50 threshold would otherwise miss by itself.

This comparative analysis serves as the final validation of the project's engineering pipeline. It creates a connection between the gap of raw algorithmic output and reliable clinical decision support.

Image Index: 00028393_000.png
Ground Truth: Infiltration



Model 1(BCE Baseline):

Pathology	Score	Threshold	Status	Note
Infiltration	%11.50	0.50	Negative	Ground Truth
Nodule	%1.09	0.50	Negative	
Consolidation	%0.32	0.50	Negative	
Pneumonia	%0.20	0.50	Negative	
Atelectasis	%0.11	0.50	Negative	

Model 2(ASL + Raw Images):

Pathology	Score	Threshold	Status	Note
Infiltration	%49.28	0.50	Negative	Ground Truth
Pleural_Thickening	%21.98	0.50	Negative	
Nodule	%21.26	0.50	Negative	
Pneumothorax	%20.08	0.50	Negative	
Fibrosis	%19.56	0.50	Negative	

Model 3(ASL + Image Processing):

Pathology	Score	Threshold	Status	Note
Infiltration	%52.89	0.50	POSITIVE	Ground Truth
Consolidation	%31.84	0.50	Negative	
Nodule	%27.97	0.50	Negative	
Pneumonia	%25.24	0.50	Negative	
Mass	%24.90	0.50	Negative	

Model 4(ASL + Class-Specific Thresholds):

Pathology	Score	Threshold	Status	Note
Infiltration	%58.18	0.50	POSITIVE	Ground Truth
Nodule	%31.16	0.55	Negative	
Mass	%27.67	0.65	Negative	
Effusion	%23.80	0.55	Negative	
Consolidation	%23.47	0.50	Negative	

Image Index: 00001317_000.png
Ground Truth: Infiltration, Pneumonia



Model 1(BCE Baseline):

Pathology	Score	Threshold	Status	Note
Consolidation	%55.21	0.50	POSITIVE	
Infiltration	%28.69	0.50	Negative	Ground Truth
Pleural_Thickening	%15.57	0.50	Negative	
Atelectasis	%15.30	0.50	Negative	
Pneumonia	%6.05	0.50	Negative	Ground Truth

Model 2(ASL + Raw Images):

Pathology	Score	Threshold	Status	Note
Infiltration	%79.87	0.50	POSITIVE	Ground Truth
Mass	%49.83	0.50	Negative	
Consolidation	%49.20	0.50	Negative	
Pneumothorax	%48.54	0.50	Negative	
Fibrosis	%44.94	0.50	Negative	

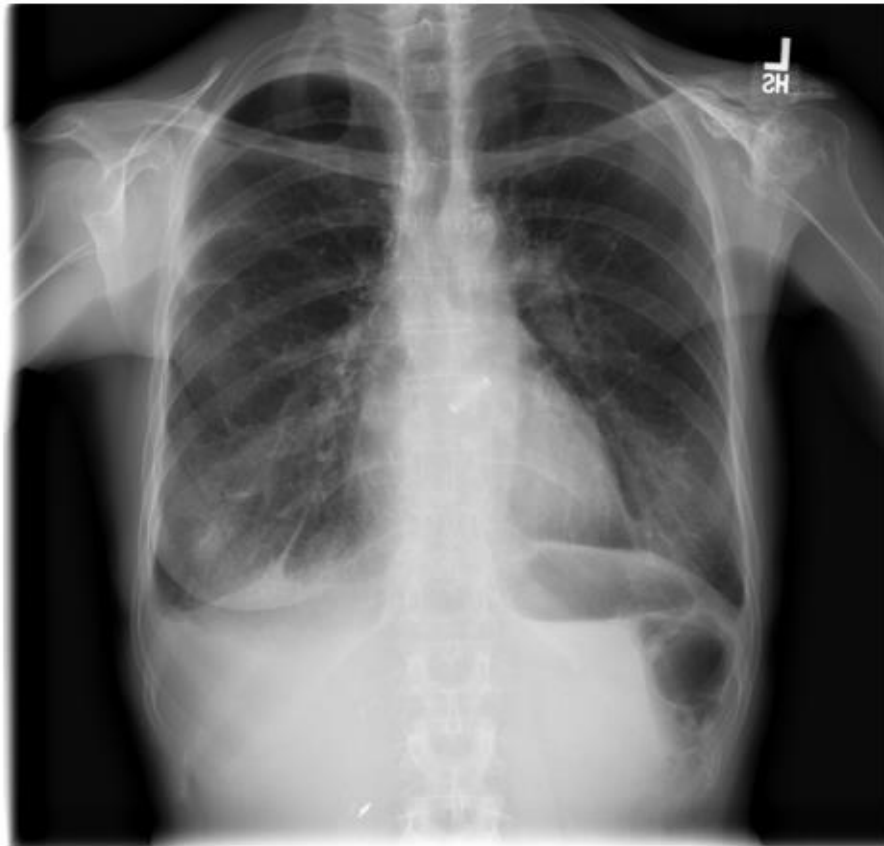
Model 3(ASL + Image Processing):

Pathology	Score	Threshold	Status	Note
Infiltration	%54.96	0.50	POSITIVE	Ground Truth
Consolidation	%43.97	0.50	Negative	
Pneumonia	%42.16	0.50	Negative	Ground Truth
Fibrosis	%39.28	0.50	Negative	
Pleural_Thickening	%37.92	0.50	Negative	

Model 4(ASL + Class-Specific Thresholds):

Pathology	Score	Threshold	Status	Note
Infiltration	%72.73	0.50	POSITIVE	Ground Truth
Consolidation	%60.25	0.50	POSITIVE	
Fibrosis	%51.80	0.50	POSITIVE	
Pneumonia	%43.39	0.35	POSITIVE	Ground Truth
Pleural_Thickening	%40.15	0.55	Negative	

Image Index: 00000092_001.png
Ground Truth: Effusion, Fibrosis



Model 1(BCE Baseline):

Pathology	Score	Threshold	Status	Note
Fibrosis	%34.43	0.50	Negative	Ground Truth
Effusion	%24.12	0.50	Negative	Ground Truth
Infiltration	%16.52	0.50	Negative	
Pleural_Thickening	%6.04	0.50	Negative	
Consolidation	%3.27	0.50	Negative	

Model 2(ASL + Raw Images):

Pathology	Score	Threshold	Status	Note
Infiltration	%57.87	0.50	POSITIVE	
Effusion	%56.34	0.50	POSITIVE	Ground Truth
Pleural_Thickening	%48.28	0.50	Negative	
Pneumothorax	%46.04	0.50	Negative	
Fibrosis	%45.02	0.50	Negative	Ground Truth

Model 3(ASL + Image Processing):

Pathology	Score	Threshold	Status	Note
Emphysema	%54.68	0.50	POSITIVE	
Fibrosis	%53.55	0.50	POSITIVE	Ground Truth
Pleural_Thickening	%48.58	0.50	Negative	
Infiltration	%47.37	0.50	Negative	
Effusion	%42.17	0.50	Negative	Ground Truth

Model 4(ASL + Class-Specific Thresholds):

Pathology	Score	Threshold	Status	Note
Pleural_Thickening	%62.46	0.55	POSITIVE	
Effusion	%61.63	0.55	POSITIVE	Ground Truth
Fibrosis	%55.73	0.50	POSITIVE	Ground Truth
Infiltration	%54.20	0.50	POSITIVE	
Pneumothorax	%50.30	0.55	Negative	

8. Conclusion of Experimental Results

The comparative analysis across all model iterations leads to the following conclusions:

- **Model 1 (BCE) Limitation:** While providing high confidence in easy cases, this model failed to detect subtle pathologies. It often produced scores below 15% for confirmed ground truth labels. This makes it unreliable for complex diagnosis..
- **The Power of ASL (Models 2 & 4):** Shifting to Asymmetric Loss boosted probability scores for rare diseases into the 45% - 65% range. This improvement made these conditions visible to the network.
- **Preprocessing Trade-off (Model 3):** Enhancements like CLAHE and Gamma Correction provided better visual contrast but introduced noise. This noise occasionally lowered the precision of the automated classifier, showing that raw images are more reliable for this architecture.
- **The Final Breakthrough (Model 4):** Model 4 emerged as the most reliable diagnostic tool. By utilizing Class-Specific Thresholds, such as 0.35 for Pneumonia, it correctly identified True Positive cases that all other models dismissed as Negative due to the rigid 0.50 threshold.

Final Verdict: Model 4 successfully bridges the gap between raw algorithmic output and reliable clinical decision support. By combining Asymmetric Loss with optimized thresholds, it achieves the highest Macro F1 score and the best overall diagnostic accuracy in the project.